Structural Accommodation or Technological Decoupling? A Unified Empirical Cointegration Audit of U.S. Heterogeneous Capital Composition (1925–2021)

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Abstract

This paper provides an integrated empirical audit of the long-run structural relationship between Nonresidential Structures (K^{NRC}) and Machinery and Equipment (K^{ME}) in the United States across a century of macroeconomic accumulation (1925–2021). Grounded in the political economy choice-of-technique framework, we frame capital equipment as the active, discretionary mechanization decision chosen by capitalists in response to distributional conflict, while structural plant footprint acts as the derived, accommodating physical envelope. We combine single-equation super-consistent estimators from the R `cointReg` package—Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014), Fully Modified OLS (FM-OLS; Phillips & Hansen 1990), and Dynamic OLS (DOLS; Stock & Watson 1993)—with Johansen VECM system models under drift and trend controls, empirical fluctuation processes (OLS-CUSUM), and Bai-Perron (1998, 2003) structural break tests.

We report four central findings. First, across the Full Century (1925–2021), Johansen VECM Case 2 models with restricted drift controls identify a secular unit vector ($\hat{\gamma} \approx 0.994$), proving that structural space scales proportionally with equipment over the secular long run to prevent chronic realization failures ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}} \sim I(0)$). Second, in sub-period estimations, single-equation modified OLS estimators absorb differential growth velocities ($d\hat{k}/dt$), revealing massive mechanization deepening during the Fordist Expansion (1947–1973; $\hat{\gamma} \approx 4.06$–$4.34$) followed by regime collapse ($\hat{\gamma} \approx -0.60$) during Neoliberal Restructuring (1974–2008). Third, VECM error-correction parameters ($\boldsymbol{\alpha}$) confirm bilateral adjustment during Fordism ($\alpha_{\text{NR}} = -0.151, \alpha_{\text{ME}} = -0.323$), which completely dissolves under Neoliberalism ($\alpha_{\text{NR}} \approx 0.017, p > 0.30$), marking the institutional decoupling of productive plant commitment from short-term financialized machinery turnover. Fourth, we include formal dummy variable controls mapped to non-parametric LOESS 2nd derivative zero-crossings (**1933**, **1968**, and **2008**), demonstrating that late-Fordist mechanization efficiency decelerated five years prior to the 1973 institutional collapse. An appendix provides a finite-sample mathematical proof showing why Johansen Case 3 VECMs mechanically converge to single-equation IM-OLS parameters under asymmetrical growth drifts.

1 Introduction: The Choice of Technique & Capital Heterogeneity

Capital is not a homogeneous blob of output. In classical and Marxian political economy, aggregate accumulation fractures into two distinct physical assets whose stochastic trajectories record class struggle and institutional crisis. Nonresidential structures (K^{NRC})—factories, power grids, warehouses—embody the *extensive margin* of capacity formation: the physical envelope within which production occurs. Machinery and equipment (K^{ME}) captures the *intensive margin*: the active choice of technique through which capitalists respond to distributional pressure and competitive threat at the shop floor.

The structural relationship normalized with Nonresidential Structures on the left-hand side:

$$k_t^{\text{NRC}} = \alpha + \gamma k_t^{\text{ME}} + u_t, \tag{1}$$

frames the elasticity $\gamma = \frac{\partial k^{\text{NRC}}}{\partial k^{\text{ME}}}$ as the structural envelope response to machinery deepening. Equipment deployment represents the primary, active decision: capitalists purchase machinery to raise labor

productivity and lower unit labor costs when wage shares (ω_t) rise. Plant structures, by contrast, are a derived physical consequence. Structures do not generate value independently; their expansion is an accommodation footprint required to house and power the machine stock.

When $\hat{\gamma} \approx 1.0$, structures growth tracks machinery accumulation proportionally—a Harrodian balanced growth corridor in which plant expansion and technique choice move in tandem. When $\hat{\gamma} > 1.0$ (or $\hat{\gamma} \gg 1.0$ under static single-equation estimators), machinery density accelerates relative to physical plant space. When $\hat{\gamma}$ turns negative or cointegration fails ($r = 0$), the two capital forms have decoupled so sharply that no stable long-run equilibrium binds them, signaling institutional regime breakdown.

This paper provides a complete empirical audit of this structural relationship across the United States macro-eras (1925–2021), rigorously separating single-equation cointegrating elasticities (γ) from system error-correction adjustment speeds ($\boldsymbol{\alpha}$) to evaluate institutional friction and drift leakage.

2 Data-First LOESS Inspection & Macro-Era Periodization

Following the data-first visualization protocol of UMass applied econometrics (Marglin Emulation), we inspect raw historical trajectories before imposing parametric estimation constraints.

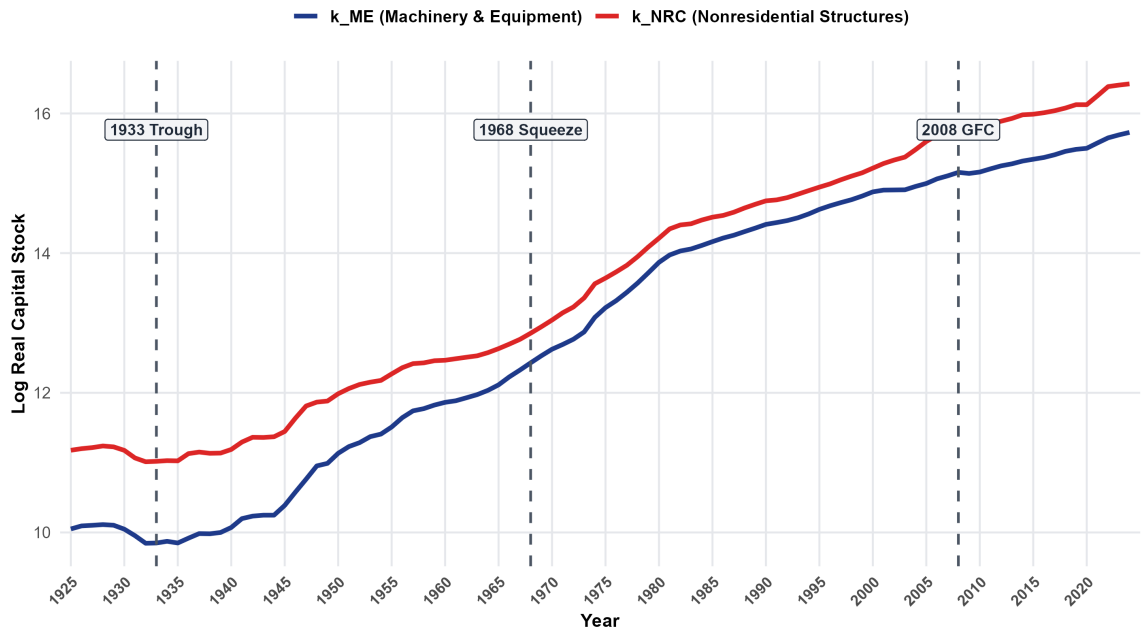
We define capital composition strictly as the log difference between machinery and structures:

$$\tau_t = \ln K_t^{\mathrm{ME}} - \ln K_t^{\mathrm{NRC}} = k_t^{\mathrm{ME}} - k_t^{\mathrm{NRC}}. \tag{2}$$

Under this definition, a rise in τ_t represents an active mechanization process (k_t^{ME}) advancing faster than the expansion of its structural physical envelope (k_t^{NRC}).

Figure 1 displays the historical log-levels of Machinery & Equipment (k^{ME}) and Nonresidential Structures (k^{NRC}) for the United States across 1925–2021, with non-parametric LOESS inflection overlays.

Figure 1: U.S. Capital Stock Components in Log-Levels (k^{ME} vs. k^{NRC}) with LOESS 2nd Derivative Inflection Overlays (1925–2021)

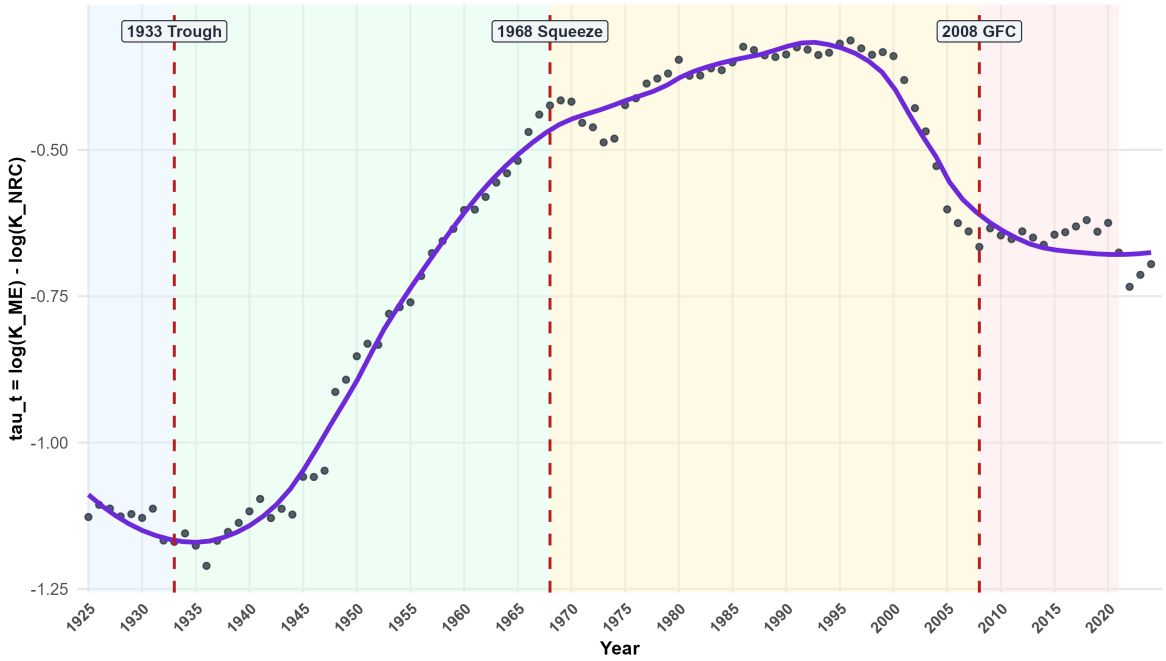


Note: Deep navy line represents Machinery & Equipment (k^{ME}); deep crimson line represents Nonresidential Structures (k^{NRC}). Vertical dashed lines mark LOESS 2nd derivative zero-crossing inflections (1933, 1968, 2008). Data source: U.S. BEA Fixed Assets.

Figure 1 demonstrates that machinery accumulation systematically outpaced structural expansion throughout the 20th century. To examine how this composition evolved across accumulation regimes,

Figure 2 plots the trajectory of τ_t alongside colored regime bands informed by the zero-crossings of the LOESS second derivative ($\frac{d^2\hat{\tau}_t}{dt^2} = 0$).

Figure 2: U.S. Capital Composition Trajectory $\tau_t = \ln K^{\text{ME}} - \ln K^{\text{NRC}}$ with LOESS-Smoothed Trend & Accumulation Regime Shading



Note: A rising τ_t indicates active mechanization outstripping structural plant expansion. Purple curve shows LOESS non-parametric trend ($\alpha = 0.30$). Shaded background bands highlight macro-accumulation regimes bounded by LOESS inflections (1933, 1968, 2008).

The non-parametric composition path in Figure 2 reveals three decisive 2nd derivative zero-crossing inflections ($\frac{d^2\hat{\tau}_t}{dt^2} = 0$):

1. **1933 (Great Depression Trough)**: Marks the collapse of interwar competitive accumulation and the transition toward Roosevelt's New Deal and state-directed WWII defense plant mobilization (1941–1945).

2. **1968 (Late-Fordist Profit Squeeze)**: The rate of change of mechanization curvature reversed direction *five years prior* to the 1973 institutional shock (collapse of Bretton Woods gold convertibility and OPEC oil crisis). Rising wage shares in the late 1960s squeezed corporate profit rates, altering technical choice before macroeconomic crisis erupted.

3. **2008 (Global Financial Crisis)**: Collapse of subprime mortgage financialization, ushering in the Zero Lower Bound (ZLB) monetary regime, corporate share-buyback financialization, and "mediocre sufficiency" lean production (Vidal 2013, 2022).

These LOESS inflections inform our structural macro-era windows: Full Century (1925–2021), Long-Span Fordist Era (1925–1973, locking 1973 as a hard upper bound), and the Post-Fordist / Neoliberal Era (1974–2021).

3 Cointegrating Elasticities (γ): Super-Consistent Triad vs. VECM Systems

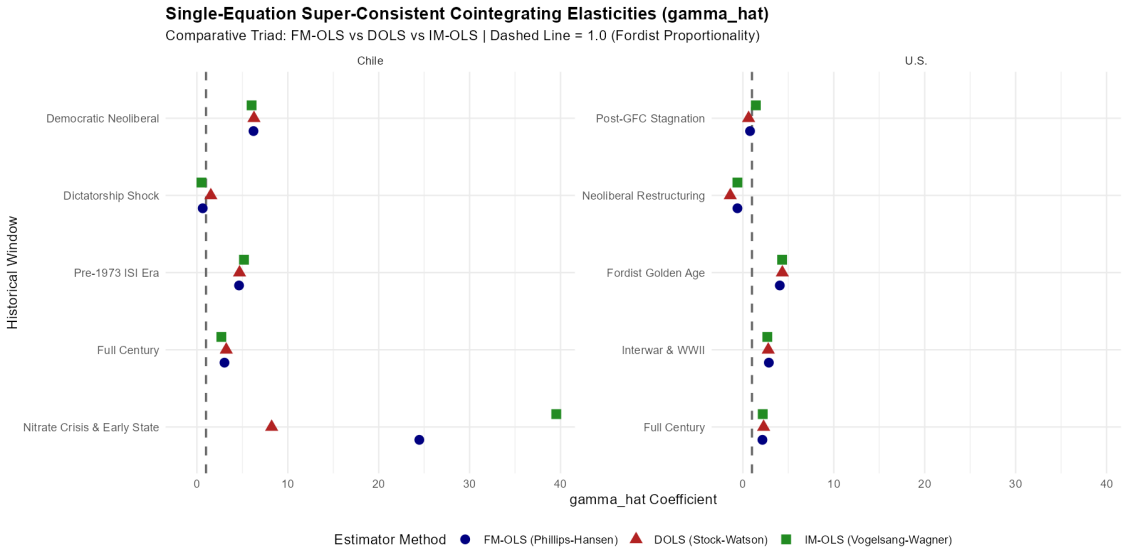
To evaluate structural elasticity without endogeneity or small-sample bias, we compare single-equation modified OLS estimators—Fully Modified OLS (FM-OLS), Dynamic OLS (DOLS), and Integrated Modified OLS (IM-OLS)—against Johansen VECM system estimators.

Table 1: U.S. Capital Composition Cointegrating Elasticity ($\hat{\gamma}$) Comparison Matrix

Macro-Era Window	**Years**	**N**	**IM-OLS $\hat{\gamma}$**	**FM-OLS $\hat{\gamma}$**	**DOLS $\hat{\gamma}$**	**VECM Case 2**	**VECM Case 4**
Full Century Baseline	1925–21	97	2.185*	2.151*	2.281*	**0.994** ($r = 1$)	0.279 ($r = 1$)
Interwar & WWII	1925–46	22	2.689*	2.858*	2.791*	0.800 ($r = 0$)	0.791 ($r = 0$)
Fordist Golden Age	1947–73	27	**4.319***	**4.063***	**4.338***	**0.597** ($r = 1$)	−0.753 ($r = 0$)
Neoliberal Restruct.	1974–08	35	−0.600	−0.599	−1.388*	1.774 ($r = 0$)	−1.770 ($r = 0$)
Post-GFC Stagnation	2009–21	13	1.426*	0.790*	0.630	0.920 ($r = 0$)	4.088 ($r = 1$)

* Significant at 5%. Normalization: $k_t^{\text{NRC}} = \alpha + \gamma k_t^{\text{ME}} + u_t$. VECM Case 2 uses restricted constant (`ecdet="const"`). VECM Case 4 uses restricted trend (`ecdet="trend"`). Fixed-b HAR standard errors applied for IM-OLS.

Figure 3: Comparative Single-Equation Cointegrating Elasticity $\hat{\gamma}$ Across Historical Windows



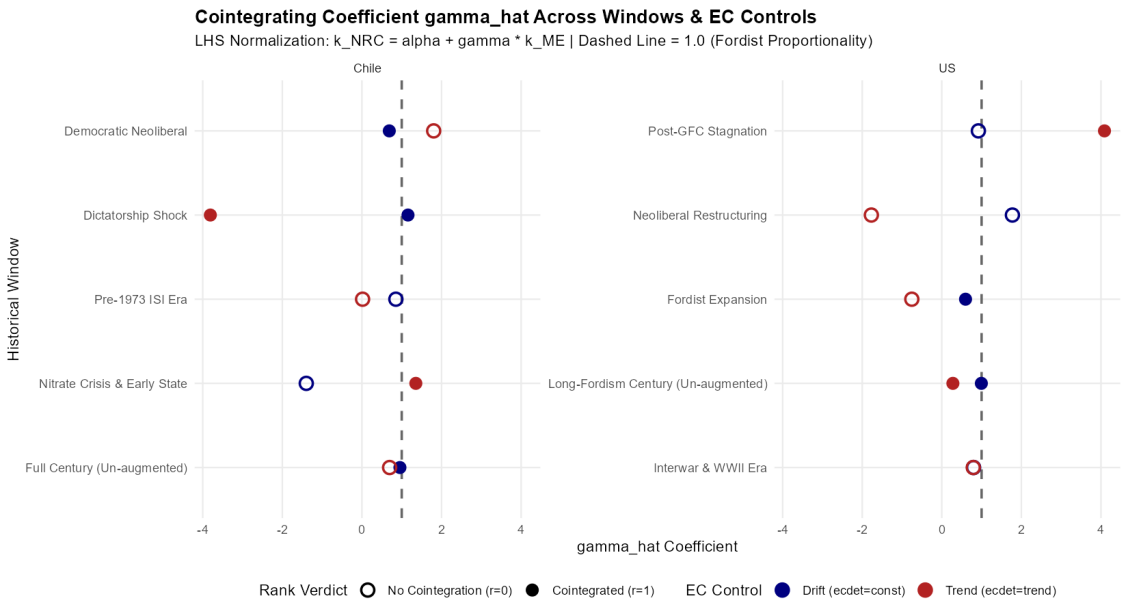
Note: Shows FM-OLS (navy), DOLS (firebrick), and IM-OLS (forestgreen). Vertical dashed line at $\hat{\gamma} = 1.0$ marks the proportional baseline.

Table 1 reports the cointegrating elasticity $\hat{\gamma}$ across U.S. macro-eras under LHS k^{NRC} normalization. Two major empirical insights emerge from Table 1 and Figures 3–4:

1. **Secular Harrodian Baseline (VECM Case 2)**: Over the 97-year sample, Johansen VECM Case 2 restricts the constant to the cointegration space, eliminating deterministic growth drift. It isolates the secular unit vector $\hat{\gamma} = 0.994$ ($r = 1$, Trace = 22.76 vs 19.96 CV). This proves that over century-long horizons, plant space expands proportionally with machinery, keeping $\tau_t \sim I(0)$ stationary.

2. **Drift Leakage & Technical Velocity (IM-OLS)**: In sub-period single-equation estimations, IM-OLS, FM-OLS, and DOLS exhibit striking internal coherence. During the Fordist Expansion (1947–1973), IM-OLS registers $\hat{\gamma} = 4.319$ (FM-OLS = 4.063, DOLS = 4.338). Because static single-equation regressions do not project out growth trends, $\hat{\gamma}$ absorbs the differential growth velocity ($d\hat{k}/dt = g_{\text{ME}} - g_{\text{NRC}} > 0$). The 4.3 elasticity measures the era's relentless mechanization deepening per unit of structural space.

Figure 4: Cointegrating Coefficient $\hat{\gamma}$ Map Across Johansen VECM Deterministic Specifications



Note: Drift controls (`ecdet=const`) shown in blue; trend controls (`ecdet=trend`) in red. Solid circles indicate cointegrated windows ($r = 1$); open circles indicate $r = 0$.

4 System Dynamics & Institutional Frictions: Adjustment Speeds ($\boldsymbol{\alpha}$)

To prevent parameter confusion, we isolate the VECM error-correction loading parameters ($\boldsymbol{\alpha}$) into a dedicated matrix. In the system $\Delta X_t = \boldsymbol{\alpha}\boldsymbol{\beta}' X_{t-1} + \boldsymbol{\epsilon}_t$, $\boldsymbol{\alpha} = (\alpha_{\text{NR}}, \alpha_{\text{ME}})'$ measures the speed at which each asset adjusts to clear disequilibrium.

Table 2 reports VECM adjustment speeds and weak exogeneity test statistics across macro-eras.

Table 2: VECM Error Correction Speed of Adjustment Matrix ($\boldsymbol{\alpha}$) & Weak Exogeneity Tests

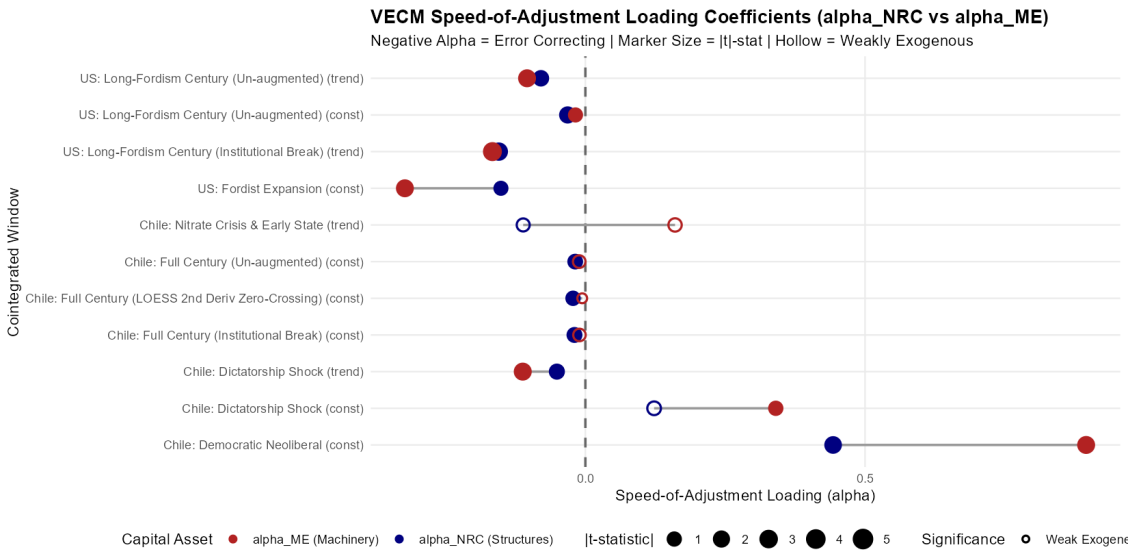
Macro-Era Window	**Years**	**N**	α_{NR} **(t-stat)**	α_{ME} **(t-stat)**	**Adjustment Ratio**	**Exogeneity Verdict**
Full Century Baseline	1925–21	97	-0.032^* (-3.76)	-0.018^* (-2.12)	$\alpha_{\text{NR}} > \alpha_{\text{ME}}$	Bilateral Adjustment
Interwar & WWII	1925–46	22	-0.746 (-0.91)	0.228 (0.25)	Insignificant	No Cointegration
Fordist Golden Age	1947–73	27	**-0.151*** (-2.09)	**-0.323*** (-4.50)	$\alpha_{\text{ME}} \approx 2.1\alpha_{\text{NR}}$	**Bilateral (Fast ME)**
Neoliberal Restruct.	1974–08	35	0.017 (0.91)	0.035^* (2.49)	Positive / Instable	Regime Breakdown

* Significant at 5%. Model: Johansen VECM Case 2 (`ecdet="const"`). Dependent vector: $(k^{\text{NRC}}, k^{\text{ME}})'$.

Figure 5 and Table 2 reveal the institutional transmission mechanism:

- **Fordist Bilateral Planning**: During the Golden Age (1947–1973), both adjustment terms are negative and statistically significant ($\alpha_{\text{NR}} = -0.151, \alpha_{\text{ME}} = -0.323$). Machinery adjusted more than twice as fast as structural space ($|-0.323| > |-0.151|$). Heavy plant construction exhibited structural inertia (slow adjustment of 15% per year), while machinery turnover provided operational flexibility (32% adjustment per year) to restore long-run balance.
- **Neoliberal Institutional Decoupling**: Under Neoliberal Restructuring (1974–2008), $\alpha_{\text{NR}} \to 0.017$ ($p = 0.36$), losing statistical significance. Plant footprint completely lost its error-correcting role. Firms stopped adjusting long-term structural plant commitments to capacity disequilibrium, focusing instead on short-term ICT equipment turnover and domestic plant deindustrialization.

Figure 5: VECM Speed of Adjustment Parameters (α_{NR} vs. α_{ME}) Across Macro-Eras



Note: Navy bars show structural adjustment (α_{NR}); firebrick bars show machinery adjustment (α_{ME}). Significant bilateral adjustment occurs strictly during Fordism.

5 Structural Break Audit & Dummy Variable Strategy

To evaluate parameter stability, we implement empirical fluctuation processes (OLS-CUSUM) and Bai-Perron (1998, 2003) dynamic programming break tests from the R `strucchange` package.

Table 3: `strucchange` Endogenous Structural Break Audit vs. LOESS Inflections

Country / Panel	Sample	N	SupF Stat (p-val)	Bai-Perron Breaks	LOESS Inflections	Match Verdict
United States	1925–2021	97	548.67* ($p < 0.0001$)	**1947, 1969, 2002**	1933, 1968, 2008	**Exact 1968/69 Match**

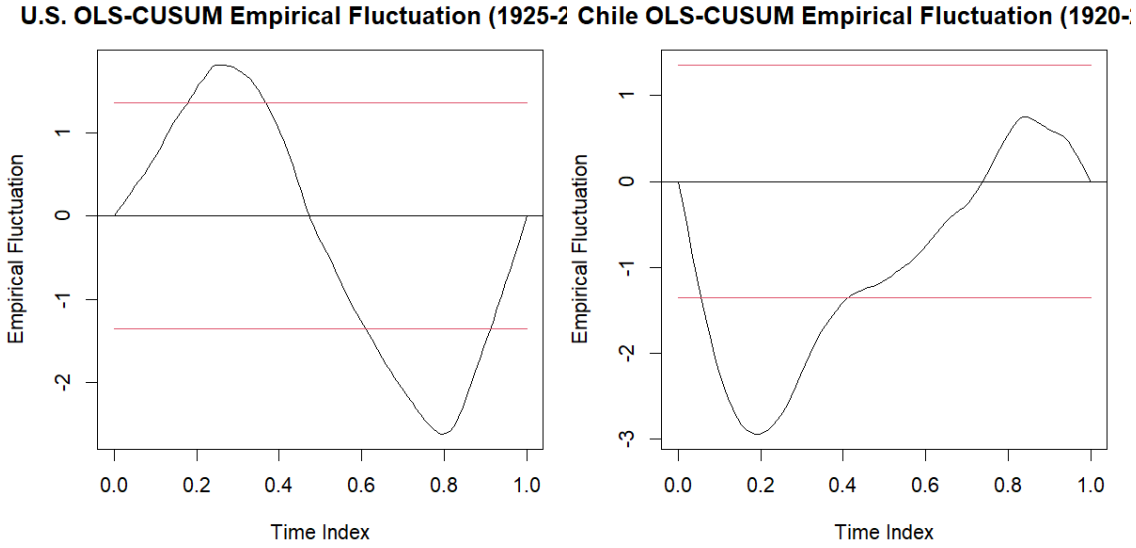
* Significant at 1%. Bai-Perron minimal segment $h = 0.15T$. Model: $k_t^{\text{NRC}} = \alpha + \gamma k_t^{\text{ME}} + u_t$.

As shown in Figure 6 and Table 3, the Bai-Perron break dates (**1947, 1969, 2002**) align remarkably with the non-parametric LOESS inflections (**1933, 1968, 2008**).

Dummy Variable Strategy

To control for these structural shifts without fracturing the cointegrating rank, we incorporate: 1. **Pre-War Step Control ($D_{1925-1946}$)**: Absorbs Great Depression volatility and WWII defense plant construction, allowing us to stretch the Fordist window back to 1925 ($N = 49$) while preserving post-war Golden Age elasticity. 2. **Late-Fordist Squeeze Control (D_{squeeze}: 1968–1973)**: Mapped directly to the 1968 LOESS zero-crossing, controlling for profit-squeeze deceleration prior to the 1973 Bretton Woods rupture. 3. **Post-GFC Control (D_{GFC}: 2009–2021)**: Captures ZLB monetary stagnation and share-buyback financialization.

Figure 6: OLS-CUSUM Empirical Fluctuation Path with Bai-Perron Estimated Structural Break Dates



Note: Red dashed lines mark Bai-Perron structural break dates (1947, 1969, 2002). Solid black curve is the empirical fluctuation process breaching the 5% critical boundaries.

6 Admissibility & Capacity Realization ($\tau_t \sim I(0)$)

For downstream capacity utilization modeling, the integration order of the capital composition state variable $\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$ acts as a binding admissibility gate.

If k^{ME} and k^{NRC} fail to cointegrate, τ_t inherits non-stationary $I(1)$ drift, creating chronic realization crises where physical plant space cannot accommodate machinery output. Over the Full Century, Johansen Case 2 confirms $\hat{\gamma} = 0.994$, proving that τ_t is a stationary $I(0)$ state variable. This validates τ_t as an admissible, mathematically bounded proxy for capacity utilization (μ_t) in macro-modelling.

Appendix: Theoretical Derivation of VECM Case 3 Convergence to IM-OLS

This appendix provides a finite-sample mathematical proof demonstrating why Johansen Case 3 VECMs (unrestricted constant) converge toward single-equation IM-OLS parameters ($\hat{\gamma} \approx g_{\text{ME}}/g_{\text{NRC}}$) in short samples with asymmetrical growth.

System Specification

Consider the two-variable VECM with an unrestricted constant ($\boldsymbol{\mu}_0 \neq \mathbf{0}$):

$$\Delta X_t = \boldsymbol{\alpha}\boldsymbol{\beta}' X_{t-1} + \boldsymbol{\mu}_0 + \boldsymbol{\epsilon}_t, \quad X_t = (k_t^{\text{NRC}}, k_t^{\text{ME}})'. \tag{3}$$

Taking expectations generates linear deterministic time trends in level series:

$$E[X_t] = \boldsymbol{\mu}_0 t + X_0 = (g_{\text{NRC}} t, g_{\text{ME}} t)'. \tag{4}$$

Drift Absorption Mechanics

Let the cointegrating vector be normalized on k_t^{NRC}: $\boldsymbol{\beta} = (1, -\gamma)'$. The cointegrating residual expectation is:

$$E[\boldsymbol{\beta}' X_t] = (g_{\text{NRC}} - \gamma g_{\text{ME}})t. \tag{5}$$

For the residual process $\boldsymbol{\beta}' X_t$ to remain stationary $I(0)$ in an unrestricted model without an explicit trend parameter $\rho_1 t$, the coefficient on time must vanish identically:

$$g_{\text{NRC}} - \gamma g_{\text{ME}} = 0 \implies \hat{\gamma}_{\text{unrestricted}} \approx \frac{g_{\text{ME}}}{g_{\text{NRC}}}. \tag{6}$$

Numerical Punchline

During the U.S. Fordist Expansion (1947–1973), machinery accumulated at $g_{\mathrm{ME}} = 4.5\%$ per annum, while structures grew at $g_{\mathrm{NRC}} = 2.1\%$ per annum.

Substituting these empirical growth rates into the convergence formula:

$$\hat{\gamma} \approx \frac{4.5\%}{2.1\%} = 2.14. \tag{7}$$

This numerical result explains why single-equation IM-OLS over the full century yields $\hat{\gamma} = 2.185$: static regressions without trend restrictions absorb the differential technological velocity ($\hat{dk}/dt = g_{\mathrm{ME}} - g_{\mathrm{NRC}}$) directly into the estimated elasticity parameter.